Lab Report – EE-170L Computer Programming (Lab 02)

NAME	saifullah.
ID	725
SEMESTER	2nd
LAB 2	(programming)
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1. Objectives

The objectives of this lab are to:

• Understand the fundamental structure and components of a C++ program.

• Learn how to perform input and output operations using cin and cout.

• Identify and use common C++ data types such as int, double, char, bool, and string.

• Apply arithmetic operators to perform basic mathematical calculations.

• Develop good programming practices through proper indentation and comments.

• Implement simple programs such as a greeting program,

calculator, and temperature converter.

2. Theory

2.1 Basic Syntax

- Every C++ program begins with to enable input/output operations.
- The function is the entry point of execution.
- is used for output, while is used for input.
- Operators and are stream insertion and extraction operators.

2.2 Data Types

Data types specify the type of data a variable can store in a program.

- int → used to store whole numbers
- double → used to store decimal or floating-point numbers
- char → used to store a single character
- bool → used to represent logical values such as true or false
- string → used to store a sequence of characters or text

2.3 Arithmetic Operators

- Addition
- Subtraction
- Multiplication
- Division
- Modulus (remainder)

2.4 Best Practices

- Indentation: Indent code inside for readability.
- Comments: Use for single-line and for multi-line explanations.

3. Programs Implemented

3.1 Hello World & User Greeting

```
#include <iostream>
```

```
using namespace std;

int main () {
  cout << "Welcome to the C++ Lab!" << endl;
   string name;
   cout << "Enter your name: ";
   cin >> name;
   cout << "Hello, " << name << "! Let's start coding." << endl;
   return 0;
}
```

3.2 Simple Calculator

```
#include <iostream>
using namespace std;

int main() {
   int num1, num2;
   cout << "Enter two integers: ";
   cin >> num1 >> num2;

   int sum = num1 + num2;
   int diff = num1 - num2;
```

```
    int product = num1 * num2;
    double quotient = (double)num1 / num2;
    int remainder = num1 % num2;

    cout << "Sum: " << sum << endl;
    cout << "Difference: " << diff << endl;
    cout << "Product: " << product << endl;
    cout << "Quotient: " << quotient << endl;
    cout << "Remainder: " << remainder << endl;
    return 0;
}
```

Lab Tasks

Task 1: Variable Types and Memory Size

```
#include <iostream>
using namespace std;

int main() {
    int a = 10;
    double b = 20.5;
    char c = 'A';
    bool d = true;
```

```
    cout << "Integer: " << a << ", Size: " << sizeof(a) << " bytes" << endl;
    cout << "Double: " << b << ", Size: " << sizeof (b) << " bytes" << endl;
    cout << "Char: " << c << ", Size: " << sizeof(c) << " bytes" << endl;
    cout << "Bool: " << d << ", Size: " << sizeof(d) << " bytes" << endl;

    return 0;
}
```

Task 2: Floating-Point Variable "pi"

```
#include <iostream>
using namespace std;

int main() {
    double pi = 3.14;
    cout << "Value of pi: " << pi << endl;
    return 0;
}
```

Task 3: Celsius to Fahrenheit Conversion

```
#include <iostream>
using namespace std;
```

```
int main() {
    double celsius, fahrenheit;
    cout << "Enter temperature in Celsius: ";
    cin >> celsius;

    fahrenheit = (celsius * 9.0 / 5.0) + 32;
    cout << "Temperature in Fahrenheit: " << fahrenheit << endl;

    return 0;
}
```

4. Results

- Successfully executed programs for input/output and arithmetic operations.
- Verified correct usage of data types and operators.
- Implemented calculator and temperature conversion programs.
- Displayed memory sizes of different variable types.
- Learned how indentation and comments improve readability.

5. Conclusion

This lab introduced the **fundamentals of C++ programming**, including syntax, input/output handling, data types, and arithmetic operations. By completing the tasks, students gained practical experience in writing simple programs and applying theoretical concepts. This foundation will be essential for more advanced programming labs.